Chapitre 2
Cryptography Basics

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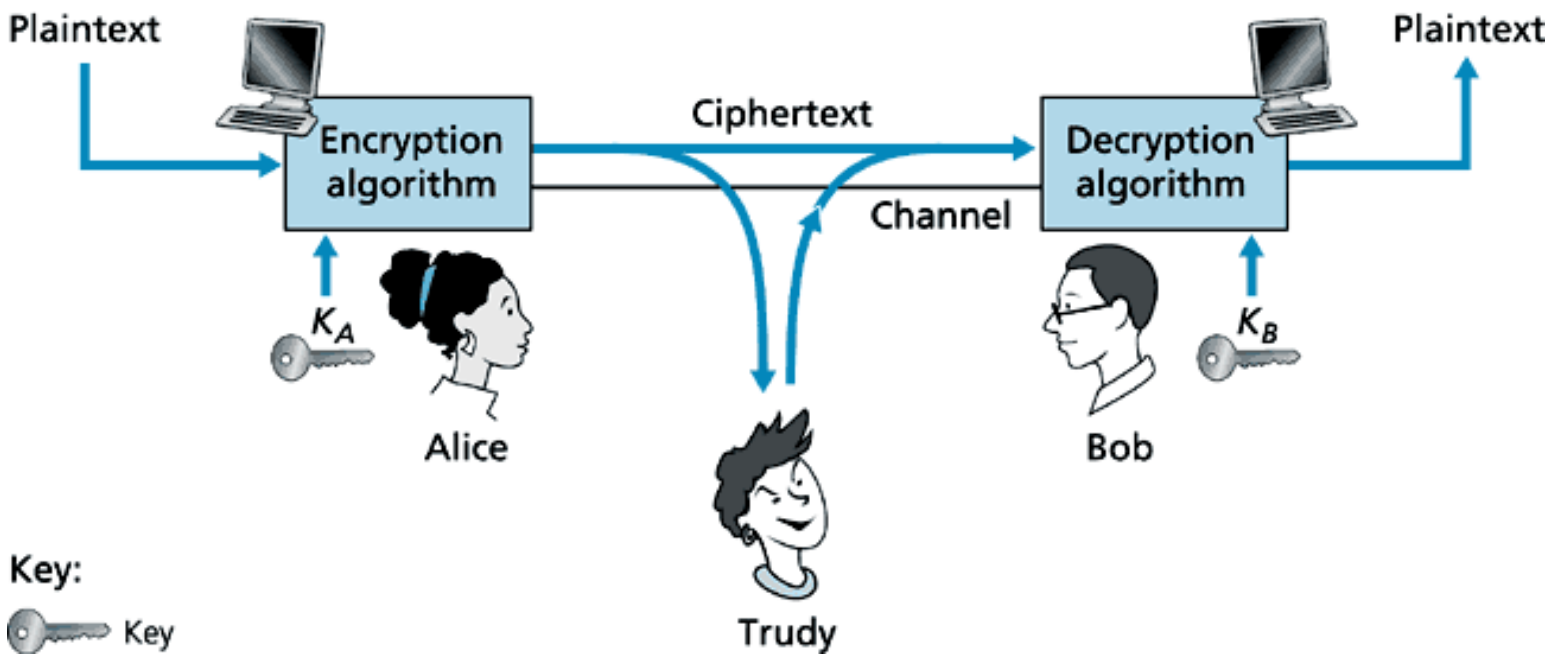
Outline

- Introduction
- Cryptography
  - Symmetric key encryption: DES, AES
  - Public key encryption: RSA, ECC
  - Hash functions
- Key distribution and certification

Introduction

Cryptography – From plaintext to cipher, and back



Introduction
Computational security (1/2)

TIME IT TAKES A HACKER TO BRUTE FORCE YOUR PASSWORD IN 2023

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	Instantly	Instantly	Instantly
6	Instantly	Instantly	Instantly	Instantly	Instantly
7	Instantly	Instantly	1 sec	2 secs	4 secs
8	Instantly	Instantly	28 secs	2 mins	5 mins
9	Instantly	3 secs	24 mins	2 hours	6 hours
10	Instantly	1 min	21 hours	5 days	2 weeks
11	Instantly	32 mins	1 month	10 months	3 years
12	1 sec	14 hours	6 years	53 years	226 years
13	5 secs	2 weeks	332 years	3k years	15k years
14	52 secs	1 year	17k years	202k years	1m years
15	9 mins	27 years	898k years	12m years	77m years
16	1 hour	713 years	46m years	779m years	5bn years
17	14 hours	18k years	2bn years	48bn years	380bn years
18	6 days	481k years	126bn years	2tn years	26tn years

HIVE SYSTEMS

› Learn how we made this table at hivesystems.io/password

- Encryption scheme is said "*computationally secure*" if
  - The cost of breaking the cipher exceeds the value of the encrypted data,
  - The time required to break the cipher exceeds the useful lifetime of the data.
- One approach to crack an encryption is a brute-force approach (*exhaustive search*)

Introduction
Computational security (2/2)

- On average, half of all possible keys have to be tried to find the needed key.
- No *security by obfuscation*
  - Use algorithms allowing public scrutiny
  - Don't design your own algorithm

Introduction
Where is the secret?

- Symmetric key cryptography
  - Sender and receiver’s keys are identical
  - They share a secret
- Public key cryptography
  - Encryption key public, decryption key secret (private)
  - Secret on single side

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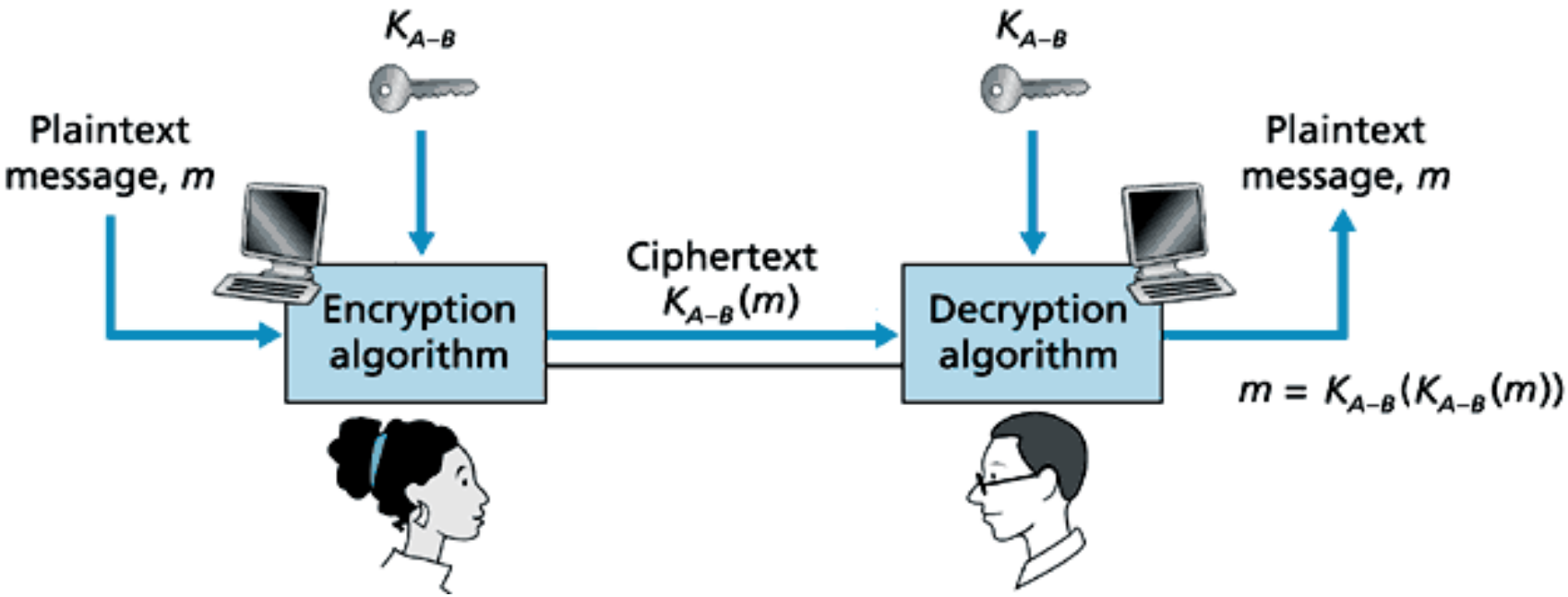
Symmetric key cryptography

- Alice and Bob share know same (symmetric) key
- Key $K_{A\text{-}B}$ is (seed for) substitution pattern
- How do Bob and Alice agree on key value?



Symmetric key cryptography
Basic examples (1/3)

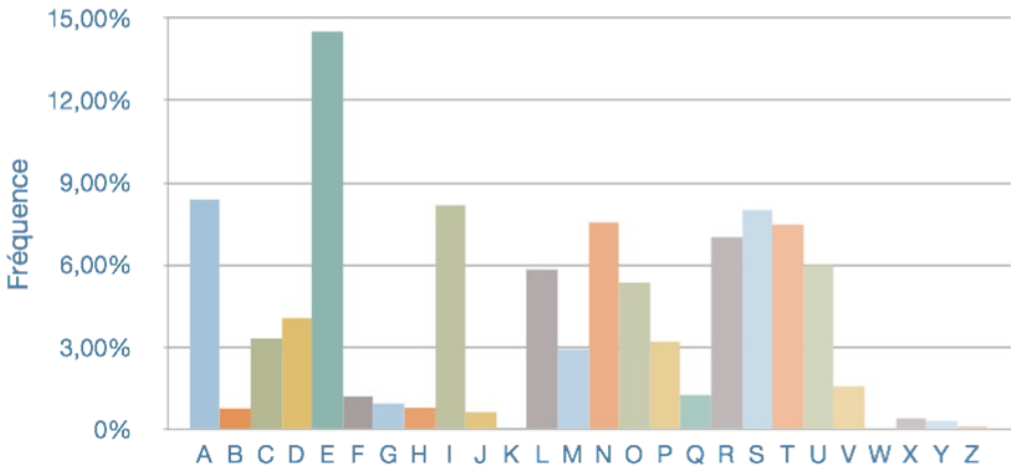
- Principle: substitute plaintext with ciphertext
- Caesar cipher *k*
  - Substitute a letter by another one, *k* positions later
  - bob, i love you. alice → ere, l oryh brx. dolfh
  - 25 key values
- Monoalphabetic cipher
  - No regular pattern

Plaintext letter:	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext letter:	m	n	b	v	c	x	z	a	s	d	f	g	h	j	k	l	p	o	i	u	y	t	r	e	w	q

  - bob, i love you. alice → nkn, s gktc wky. mgsbc
  - 26! = 403 10^{24} key values

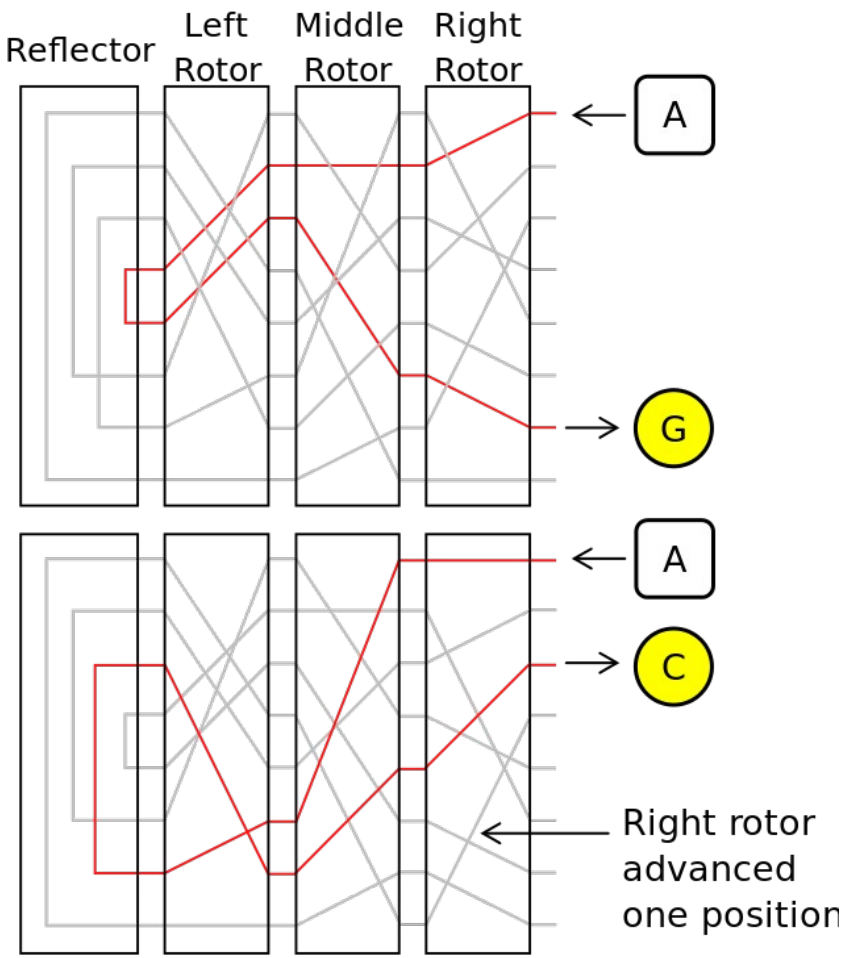
Symmetric key cryptography
Basic examples (2/3)

- Monoalphabetic cipher (con't)
  - Attack eased by statistical analysis of content



- Polyalphabetic encryption
  - Multiple, successive encodings
  - Example: Caesar ciphers k = {5, 19} + $C_1C_2C_2C_1C_2$
  - bob, i love you. alice → wvi, d sjcl tvp. hsdjz

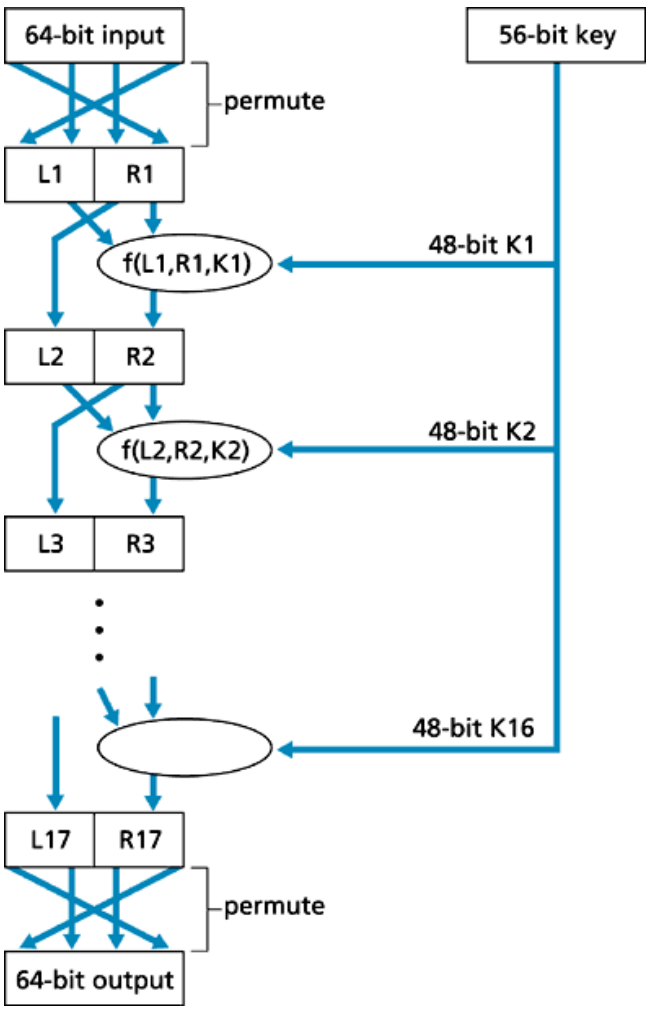
Symmetric key cryptography
Basic examples (3/3)



Source : Wikipedia

- Polyalphabetic encryption (con't)
  - Enigma (World War II)
  - Substitution scheme changed for each character
  - Secret: rotors + initial position of rotors and cables
  - Brute-force: 158,962,555,217,826,360,000 settings

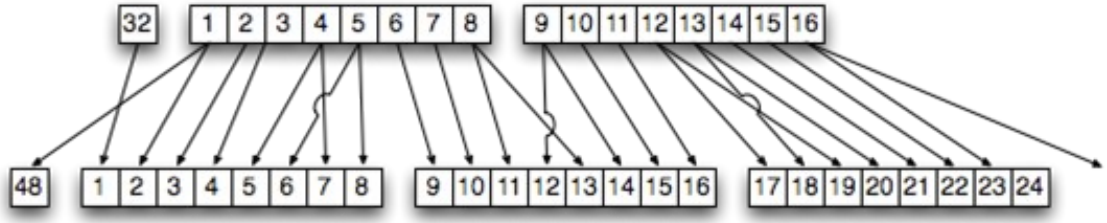
Symmetric key cryptography
Data Encryption Standard (DES, 1/3)



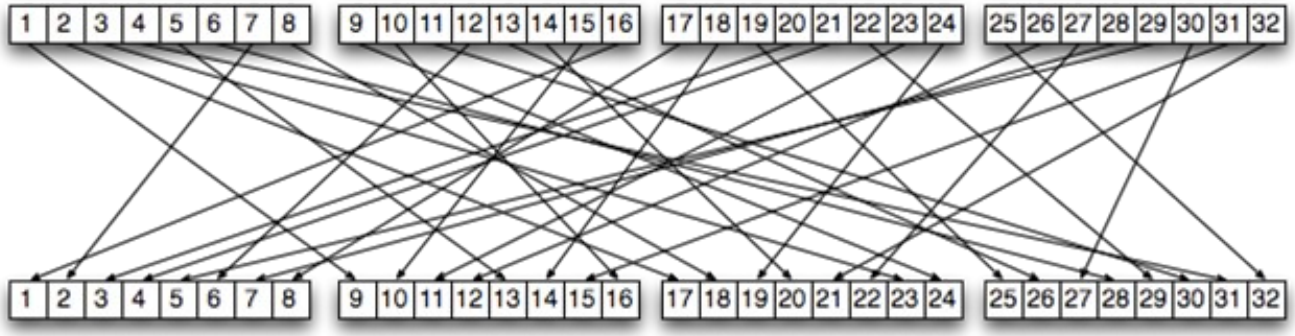
- Chunks of 64-bit plaintext input encrypted with 56-bit symmetric key
- Encryption
  - Initial permutation
  - 16 identical “rounds” of
    - Swap left-right
    - Function application, each using a different 48-bit key derived from the 56-bit key
    - Function combines expansion, substitution and XOR
  - Final permutation
- Decryption: reverse operations

Symmetric key cryptography
Data Encryption Standard (DES, 2/3)

- Expansion (E-Box)



- Permutation (P-Box)



- Actual connections depend on the secret

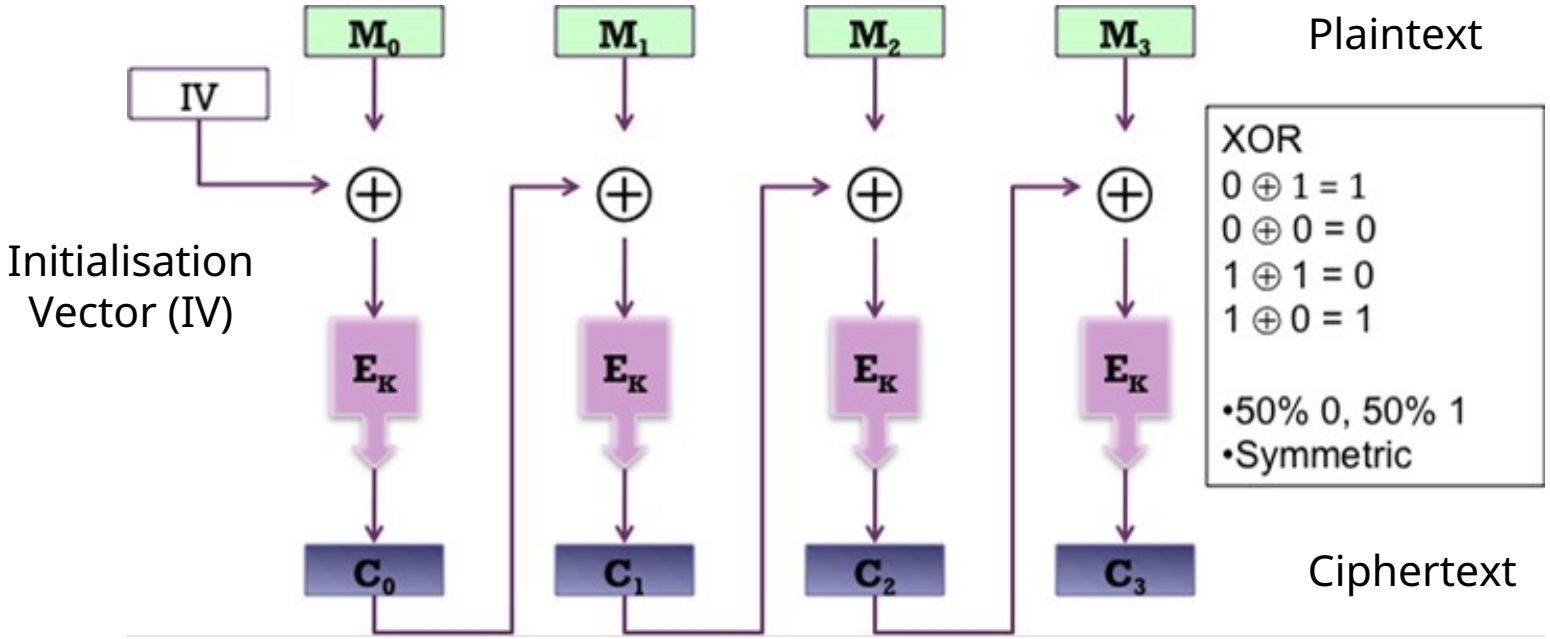
Symmetric key cryptography
Data Encryption Standard (DES, 3/3)

- Goal: no correlation between ciphertext and either original data or encryption key
- No known “backdoor” decryption approach, only exhaustive search
- How secure is DES? DES Challenge

Challenge I	January 1997	96 days
Challenge II-1	February 1998	41 days
Challenge II-2	July 1998	56 hours
Challenge III	January 1999	22,25 hours

Symmetric key cryptography
3-DES – CBC

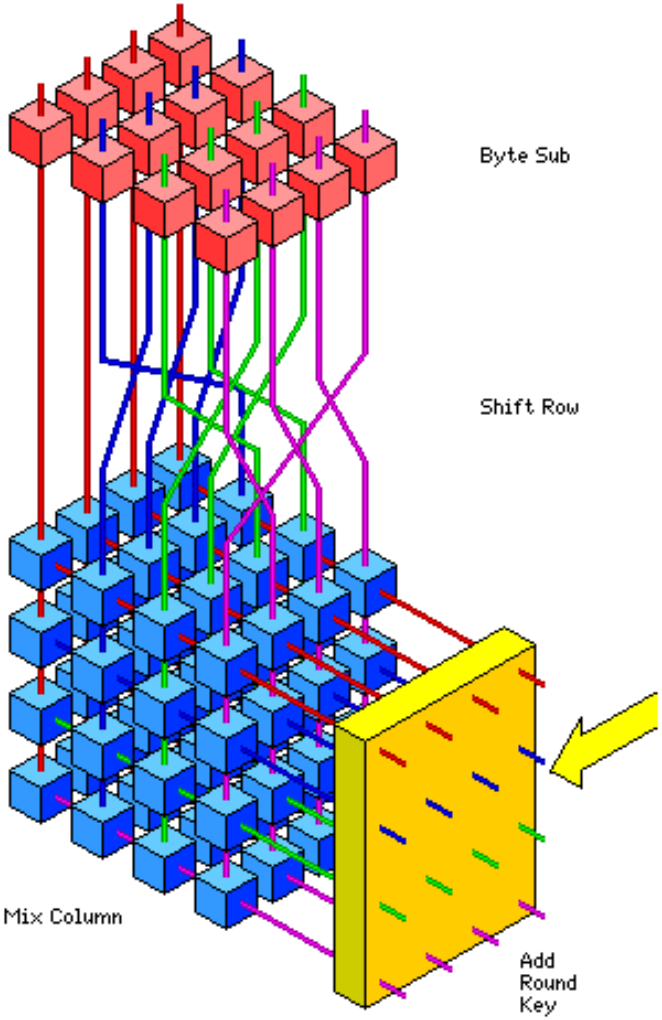
- 3-DES: sequence of three DES encodings
- Cipher-Block Chaining (CBC)
  - XOR(K(m^j), $m^{(j+1)}$) before cyphering $m^{(j+1)}$
  - IEEE 802.11i



Symmetric key cryptography
Advanced Encryption Standard (AES, 1/2)

- Originally proposed in 2000 as Rijndael by Vincent Rijmen (KULeuven) and Joan Daemen (Proton World International)
- Adopted by NIST as DES replacement
- A machine able to crack 56-bit DES in one second would need 149 10^{12} years to crack AES
- No academic attack based on cryptanalysis found yet. Repeated attempts to shorten sequence of operations.

Symmetric key cryptography
Advanced Encryption Standard (AES, 2/2)



- Chunks of 128 bits using 128-, 192- and 256-bit keys
- 10/12/14 identical “rounds” of
  - Byte substitution
  - Row shifting
  - Column mixing
  - Addition with key + rounding
- Illustration

Source: http://home.ecn.ab.ca/~jsavard/crypto/images/rijnab.gif

Symmetric key cryptography
AES cryptanalysis

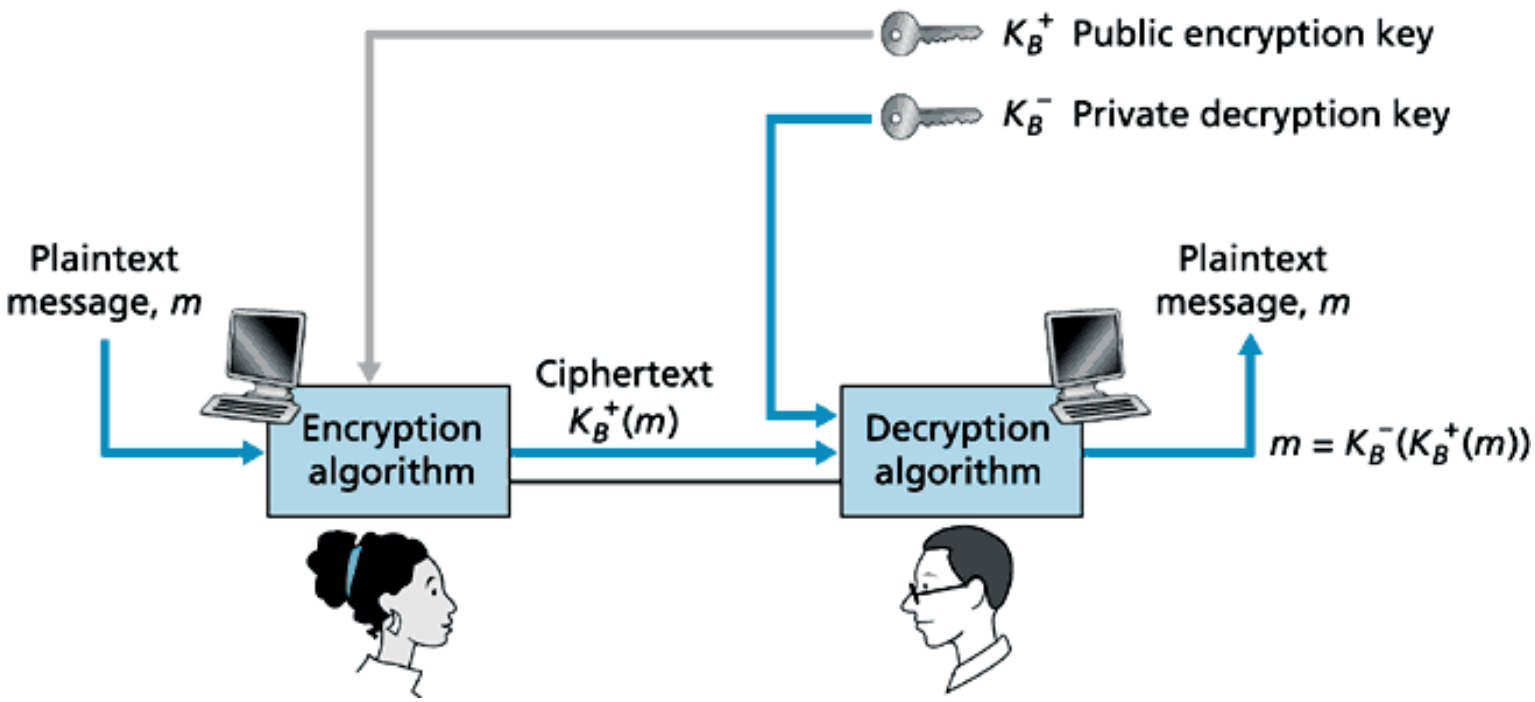
- Repeated attempts to shorten the sequence
- First meaningful succes in august 2011
- Equivalent to a 2-bit reduction of the key length (gain = 4)
- Number of steps for brute-force attacks remains $8*10^{37}$
- With 10^{12} devices testing 10^{9} keys per second, still $2*10^{9}$ years requested to find a given key

Symmetric key cryptography
Other algorithms

- Non exhaustive list
  - Ron Rivest's Ciphers RC2, RC4 (Prohibited, RFC 7465), RC5, RC6 (AES candidate)
  - International Data Encryption Algorithm (IDEA)
  - CAST (C Adams, S. Tavares)
- Similar method
  - Split plaintext in *N*-bit words
  - Repeat basic secret key-dependent operations
  - Substitution, permutation, XOR, etc

Public key cryptography

- Alice fetches Bob's public key K_B^+ and encrypts message using K_B^+ and a standardised algorithm
- Bob decrypts ciphertext using his own private key K_B^-



Public key cryptography
Requirements

- Pretty simple
- Two requirements
  - $K_B^-(K_B^+(m)) = m = K_B^+(K_B^-(m))$
  - Given the public key K_B^+, it should be impossible to derive the private key K_B^-
- Drawbacks
  - Enables chosen plaintext attacks, since anyone can send a ciphered message
  - Key ownership not longer authenticates sender
- Algorithms
  - 1st generation: RSA (Rivest, Shamir and Adleman)
  - 2nd generation: ECC (Elliptic Curve Cryptography)

Public key cryptography
RSA – Key generation

- Using modulo-n arithmetic
  - Choose two large prime numbers p, q (e.g. 1,024 bits each)
  - Compute $n = pq$, $z = (p-1)(q-1)$
  - Choose e (with $e < n$) that has no common factors with z (e, z are "relatively prime")
  - Choose d such that $ed-1$ is exactly divisible by z (in other words: $ed \bmod z = 1$)
  - Public key is $K_B^+ = (n,e)$, private key is $K_B^- = (n,d)$
- Example
  - $p = 5$, $q = 7$
  - $n = 35$, $z = 24$, $e = 5 < 35$, $d = 29$

Public key cryptography
RSA – Key generation with OpenSSL

- Generate a pair of keys

```
openssl genrsa -out my.key -aes128 1024
```

- Check the keys (actually *p*, *q*, *n*, *e*, *d*)

```
openssl rsa -in my.key -noout -text
```

- Extract public key K^+

```
openssl rsa -in my.key -out mypub.key -pubout
```

Public key cryptography
RSA – Cyphering and decyphering

- Given (n, e) and (n, d) as computed above
  - To encrypt bit pattern m, compute $c = m^e \bmod n$ (i.e., remainder when m^e is divided by n)
  - To decrypt bit pattern m, compute $c^d \bmod n$ (i.e., remainder when c^d is divided by n)

$$
\begin{aligned}
c^d \bmod n &= \left(m^e\right)^d \bmod n \\
&\overset{\Delta}{=} m^{\left[ed \bmod (p-1)(q-1)\right]} \bmod n \\
&= m^1 \bmod n
\end{aligned}
$$

- Security relies on the absence (so far) of algorithms for quickly factoring n into p and q
- Important property: $K_B^-(K_B^+(m)) = m = K_B^+(K_B^-(m))$

Public key cryptography
RSA – Example

Plaintext Letter	m: numeric representation	m^e	ciphertext $c = m^e \bmod n$
l	12	248832	17
o	15	759375	15
v	22	5153632	22
e	5	3125	10

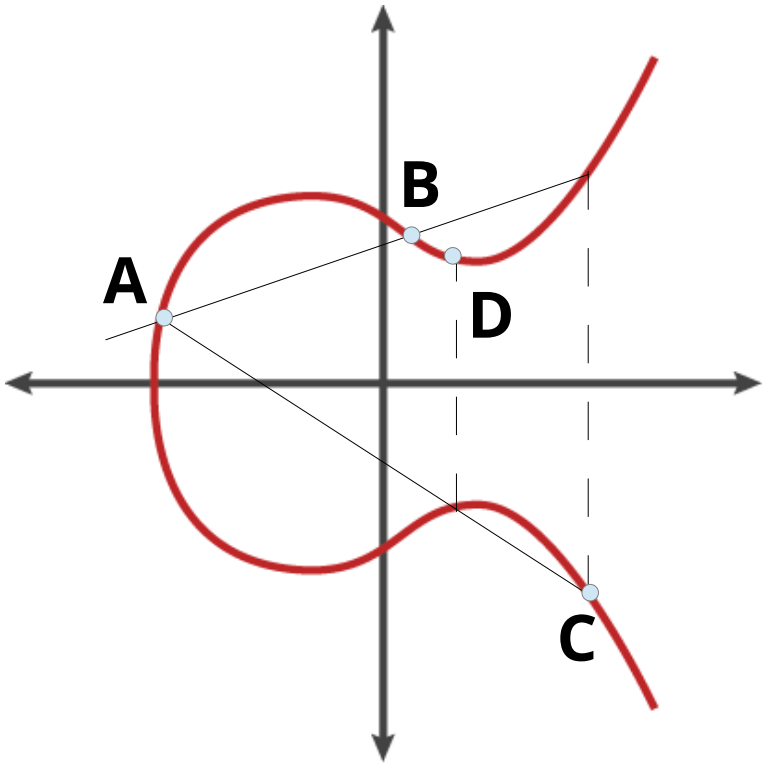
Ciphertext c	c^d	$m = c^d \bmod n$	Plaintext Letter
17	481968572106750915091411825223071697	12	l
15	12783403948858939111232757568359375	15	o
22	851643319086537701956194499721106030592	22	v
10	10000000000000000000000000000	5	e

- RSA generates extremely large numbers
- Exponentiation is computationally demanding

Public key cryptography
RSA – Limits

- Moore's Law forces the increase of the size of the prime numbers
- Two issues
  - Smartphones and similar devices miss CPU and battery
  - Factorisation approximation algorithms (Quadratic Sieve, General Number Field Sieve) compute better results for large numbers
- Cryptography base on prime number factorisation is doomed
- Five years according to MIT Technology Review (august 2013)

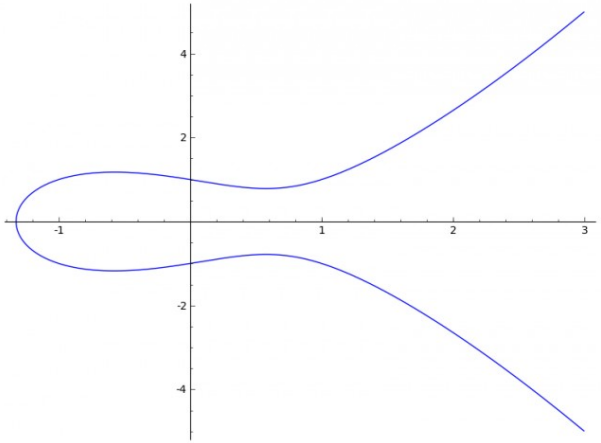
Public key cryptography
ECC – Principle (1/2)



- $y^2 = x^3 + ax + b$
- Symmetry w.r.t. x-axis
- Three section points per line
- Kind of snooker game
- Knowing the curve, point A and final point, how many runs to reach it ?

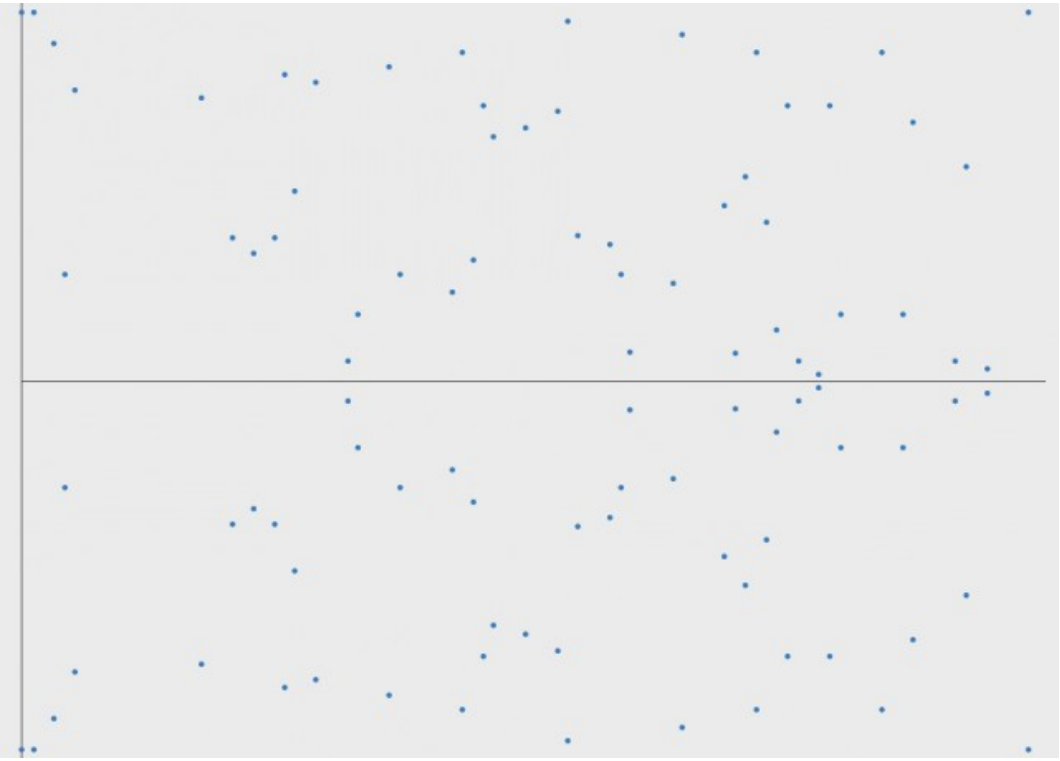
Nick Sullivan, « *A (relatively easy to understand) primer on elliptic curve cryptography* », Ars Technica, october 2013

Public key cryptography
ECC – Principle (2/2)



$y^2 = x^3 - x + 1$

Modulo 97
Integers
only



Public key cryptography
ECC vs. RSA

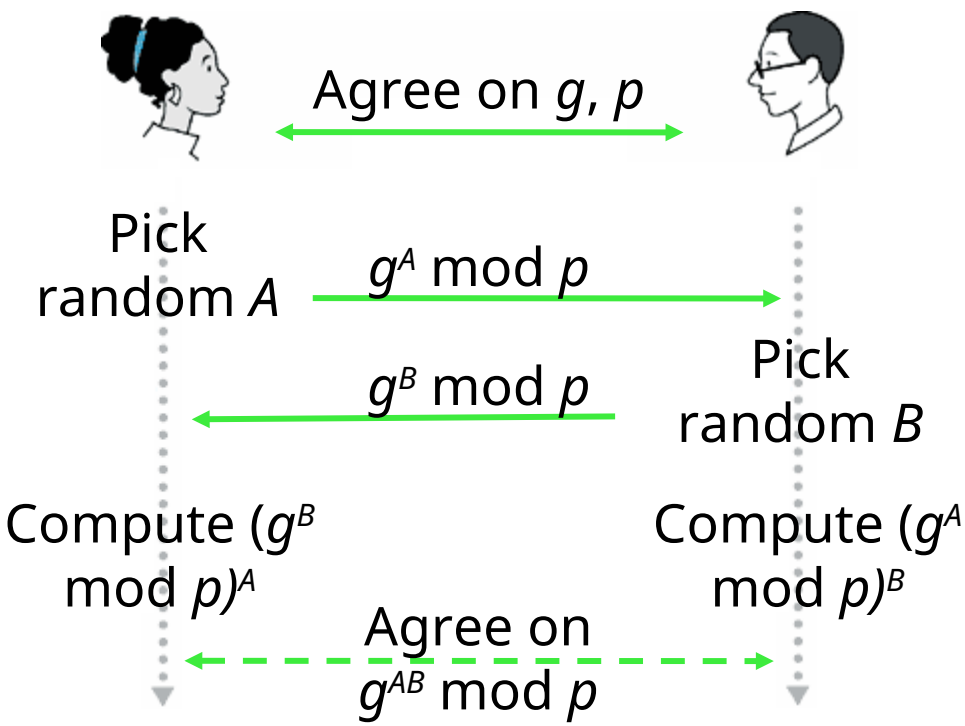
- Assuming
  - A prime number as modulo
  - An elliptic curve
  - A starting point A
- The pair of keys is
  - K^-: a number n
  - K^+: the point reached on the elliptic curve after n bounds
- Mathematical foundation : elliptic curve discrete logarithm
- Currently unsolvable but by brute force
- Same protection as RSA, but with shorter keys

Cryptography
Symmetric key vs. public key

Symmetric key	Public key
• Requires sender and receiver know shared secret key • How to agree on key in first place (particularly if never “met”)?	• Radically different approach • Sender and receiver do not share secret key • Public encryption key K^+ known to all • Private decryption key K^- known only to receiver
Based on substitutions and permutations	Based on mathematical functions

Cryptography
Session key – Diffie-Hellman or mixed scheme

- Diffie-Hellman

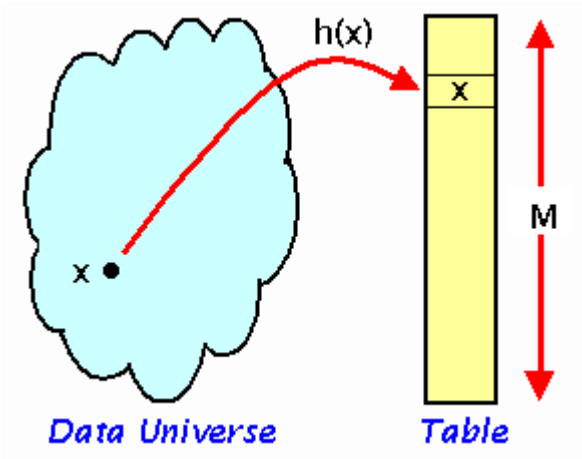


Given g, p and g^X *mod* p, guess X is tricky

- Mixed scheme
  - Alice generates an AES key K_S
  - Alice encrypts it using Bob's public key K_B^+
  - Bob receives the RSA-encrypted key and decrypts it with its private key K_B^- to obtain the AES session key K_S

Cryptography
Hash functions (1/2)

- Many-to-1 mapping
- Produces fixed-size outcome
- Examples: checksum, CRC



Message	ASCII Representation			
I O U 1	49	4F	55	31
0 0 . 9	30	30	2E	39
9 B O B	39	42	4F	42
	B2	C1	D2	AC

Checksum

Poor hash: easy to find two messages with same checksum

Message	ASCII Representation			
I O U 9	49	4F	55	39
0 0 . 1	30	30	2E	31
9 B O B	39	42	4F	42
	B2	C1	D2	AC

Checksum

Source: http://cs.engr.uky.edu/~lewis/essays/algorithms/hashing/hashing.html

Cryptography
Hash functions (2/2)

- Property of a strong hash function: it is computationally infeasible to find two messages x and y such that $H(x) = H(y)$
- Strategies
  - Multiplicative: $H(x) = 1 + (x/k)*M$ if $0 < x < k$
  - Modular: $H(x) = 1 + x \bmod M$
  - Hybrid: $H(x) = 1 + \alpha x \bmod M$
- Rules of thumb
  - M = prime number
  - $\alpha = 0.618033$

Cryptography
Hash functions – Examples

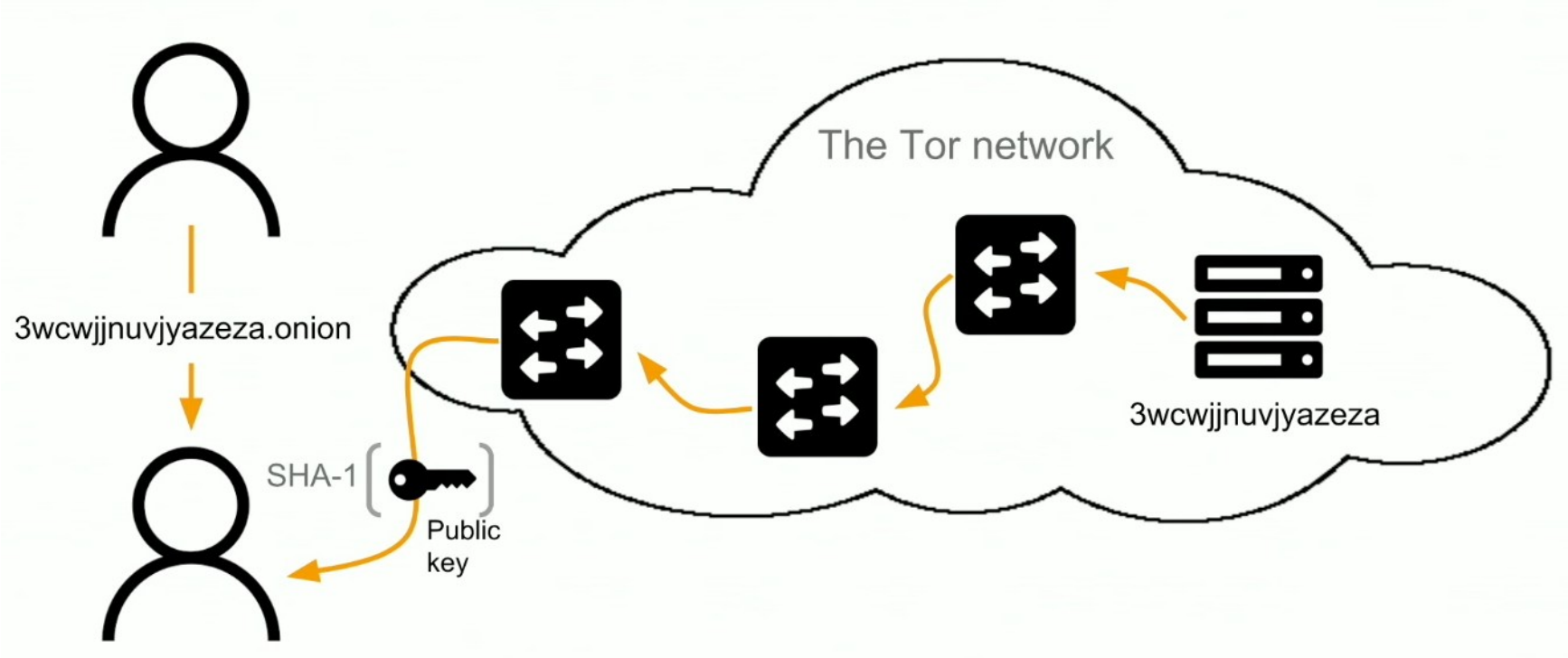
- MD2/4/5 hash functions (RFC 1321)

```
openssl dgst -md5 plaintext.txt
```

- Secure Hash Algorithm : SHA-0/1/2/3 (a.k.a. Keccak)
- RIPEMD-128/160/256/320

Cryptography
Hash functions – Onion services

- TOR network is a decentralised anonymity network
- Onion services provide server anonymity
- Onion services are self-authenticating



Source : https://youtu.be/MYR4sB3wPOg

Cryptography
Summary

Symmetric key	Public key	Hash functions
AES	ECC	MD2/4/5
ChaCha20	RSA	SHA-0†/1†/2/3
3-DES		RIPEMD-128/160
RC2, RC4†, RC5		Poly1305
IDEA		
CAST		

Cryptography
Post-Snowden era

- No hint that mathematical one-way functions have been inverted
- But hints that implementations have been deliberately weakened

Cryptography
Post-Quantum Cryptography (PQC) for Engineers

- draft-ietf-pquip-pqc-engineers (Work in progress, August 2025)
- Integer factorization and discrete logarithms happen to fall within the niche that quantum computers are expected to excel at.
- “Shor’s algorithm” efficiently solves the integer factorization problem (and the related discrete logarithm problem)
- Any asymmetric cryptographic algorithm based on integer factorization, finite field discrete logarithms, or elliptic curve discrete logarithms will be vulnerable to attacks using Shor's algorithm on a CRQC
- New kind of attack : « Harvest now, decrypt later » (HNDL)

Cryptography
Post-Quantum Cryptography (PQC) for Engineers

- What about symmetric-key cryptography?
- Grover's algorithm is a quantum search algorithm providing a theoretical *quadratic* speedup for searching an unstructured database
- Common misconception: key lengths to be *doubled*
- But Grover's algorithm is highly non-parallelizable
- Even if one deploys 2^c computational units in parallel to brute-force a key, Grover's algorithm will complete in time proportional to 2^{(128-c)/2}
- Using 256 quantum computers will only reduce runtime by a factor of 16, 1024 by a factor of 32 and so forth
- General consensus : AES-128 (Advanced Encryption Standard) remains secure in practice, and key sizes do not necessarily need to be doubled.

Cryptography
Summary

Cryptography		Until 2020	2021-2030	> 2030
Symmetric (AES)	Keys	100 bits	128 bits	128 bits
	Words	64 bits	128 bits	128 bits
Factorisation (RSA)	Modulo	2,048 bits	2,048 bits	3,072 bits
	Exponants	$\geq 2^{16}$	$\geq 2^{16}$	$\geq 2^{16}$
ECC	Modulo	200 bits	256 bits	256 bits
Hash	Digest	200 bits	256 bits	256 bits

Source : Agence Nationale de la Sécurité des Systèmes d'Information, « Mécanismes cryptographiques – Règles et recommandations, Rev. 2.03, ANSSI , 02/2014 »

Outline

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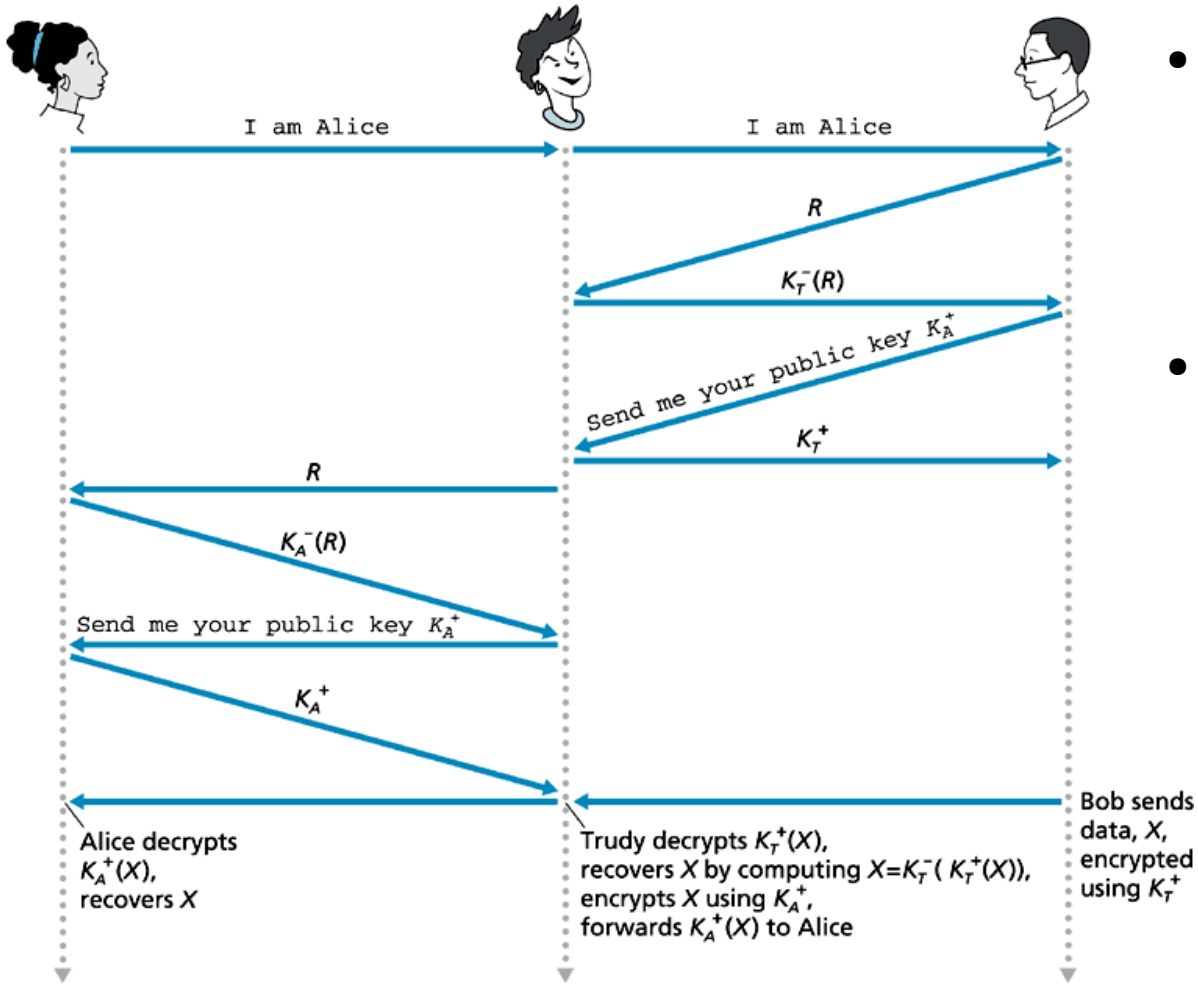
Key distribution and certification
Person-in-the-middle attack



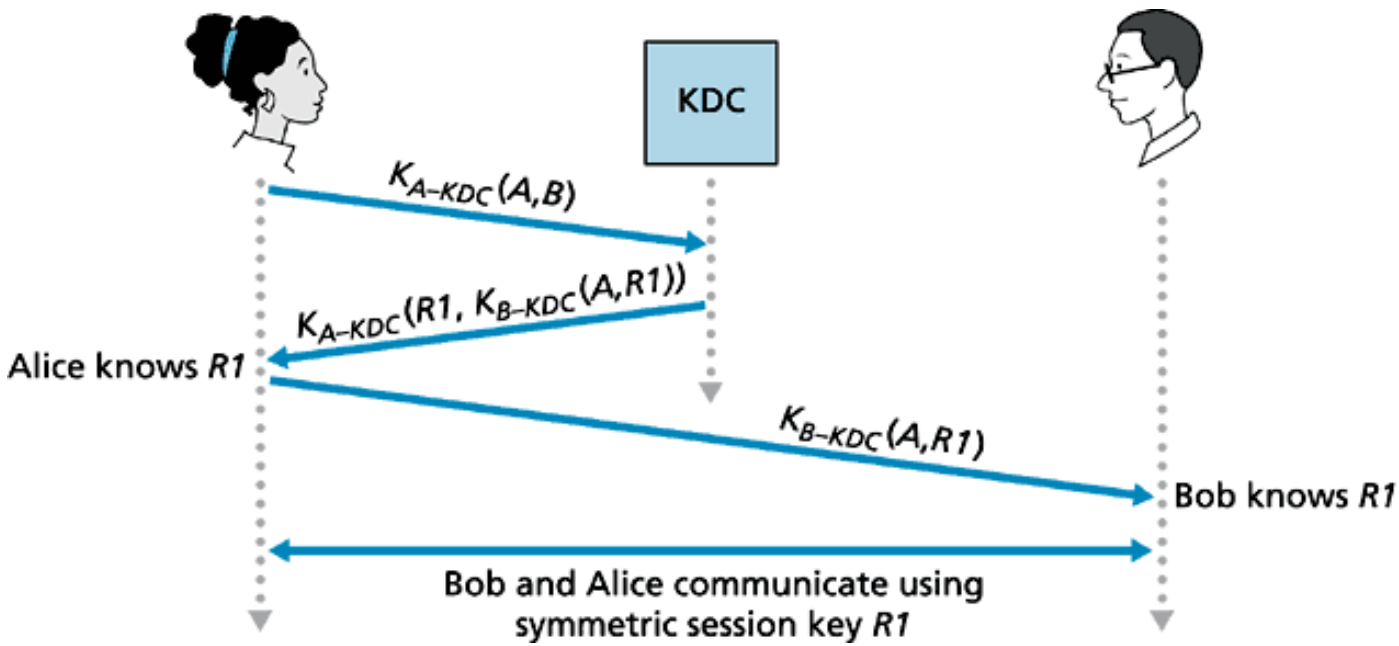
- Trudy can impersonate Alice with her own pair of keys
- Eventually, Alice and Bob will find out

Key distribution and certification
Trusted Third Party (TTP)

	Symmetric key	Public key
Issue	How to share a secret key?	How to securely collect a public key?
Solution: TTP	Key Distribution Centre (KDC)	Certification Authority (CA)

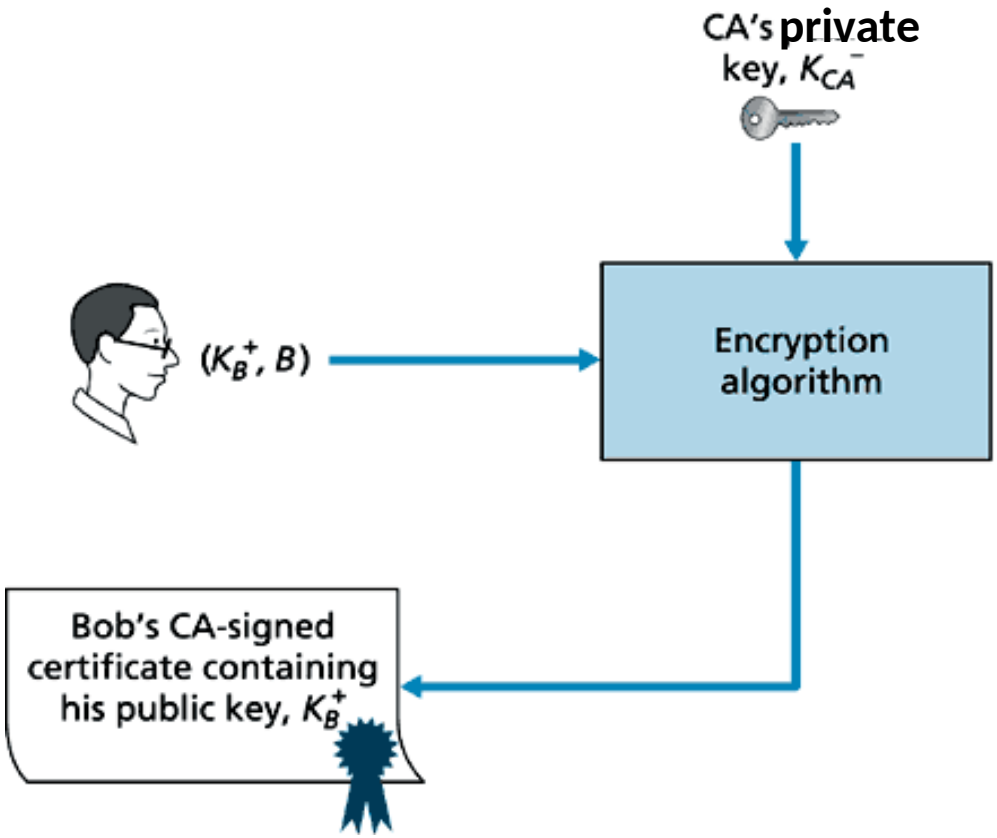
Key distribution and certification
Key Distribution Center (KDC)

- The KDC shares a secret key with each user
- A given user only knows the key it shares with the KDC
- The KDC helps its registered users to share a one-time session key (Needham-Schroeder protocol)



Key distribution and certification

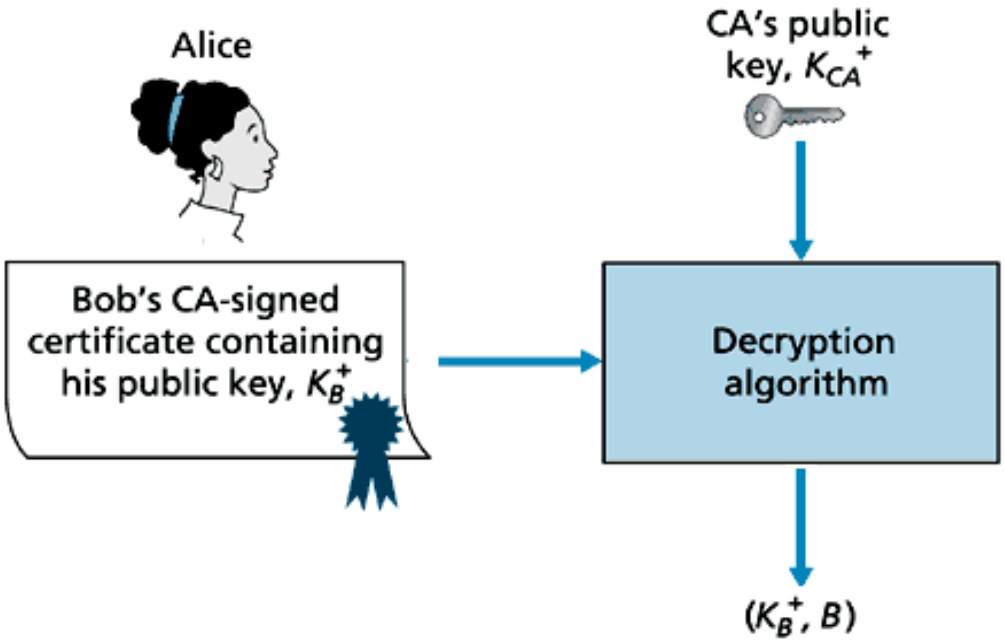
Certification Authority (CA) – X.509 standard



- B provides "proof of identity" to CA.
- CA creates certificate binding B to K_B^+
- Certificate containing B's public key digitally signed by CA
- CA says "this is B's public key"

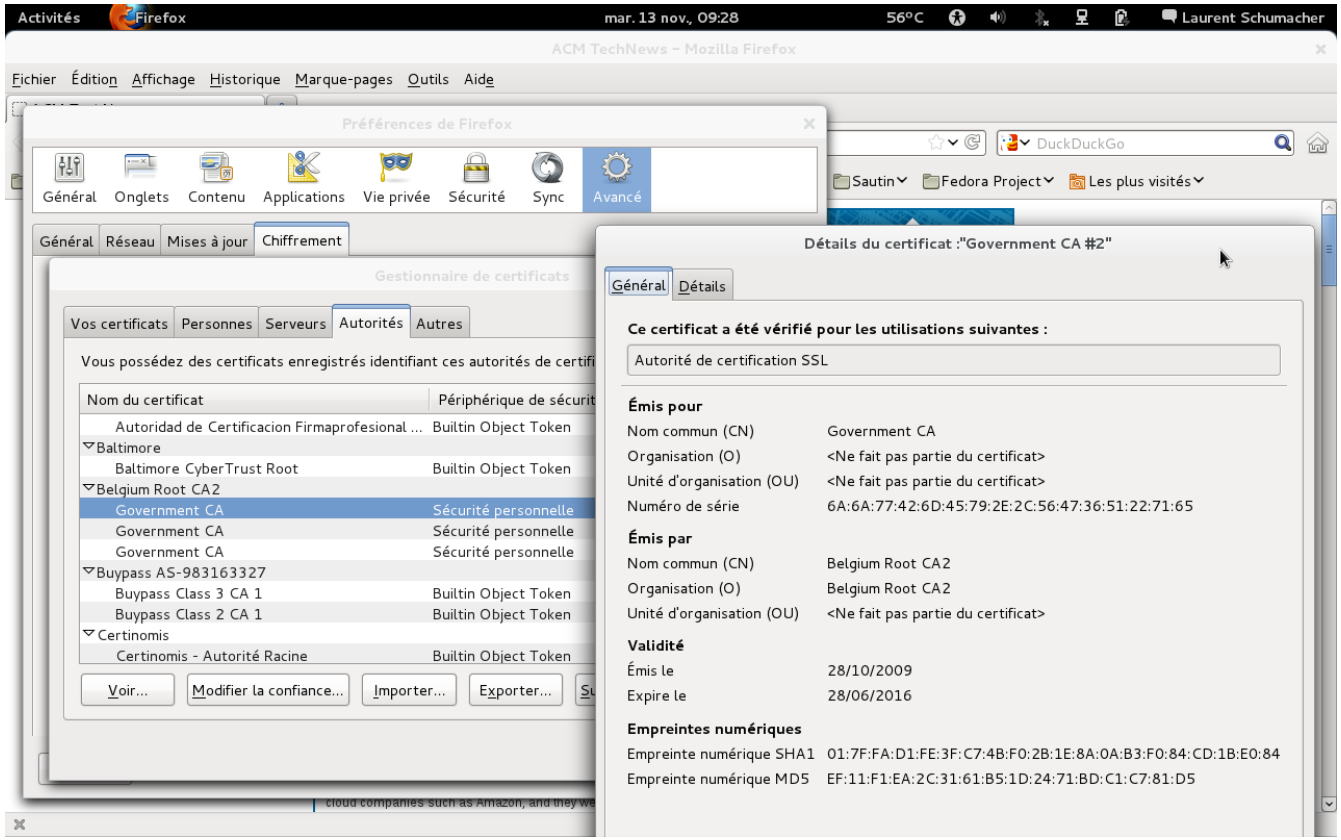
Key distribution and certification
Certification Authority (CA) – X.509 standard

- When Alice wants Bob's public key:
  - She gets Bob's certificate, either directly from Bob or elsewhere
  - She applies CA's public key to Bob's certificate to extract Bob's valid public key



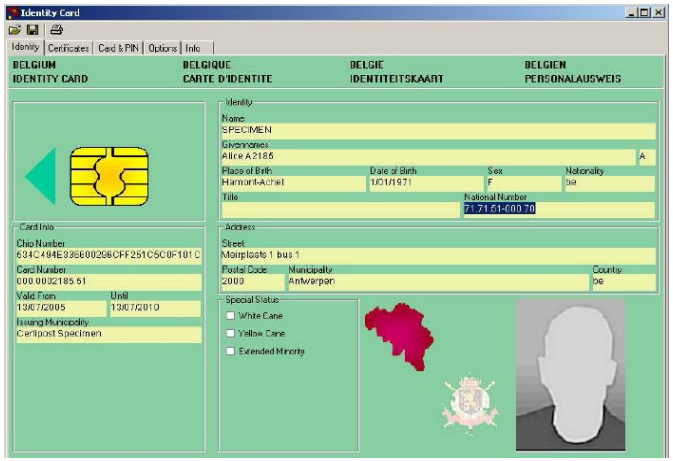
Key distribution and certification
Certification Authority (CA) – X.509 standard

- X.509 = ITU-T standard for Public Key Infrastructure (PKI)



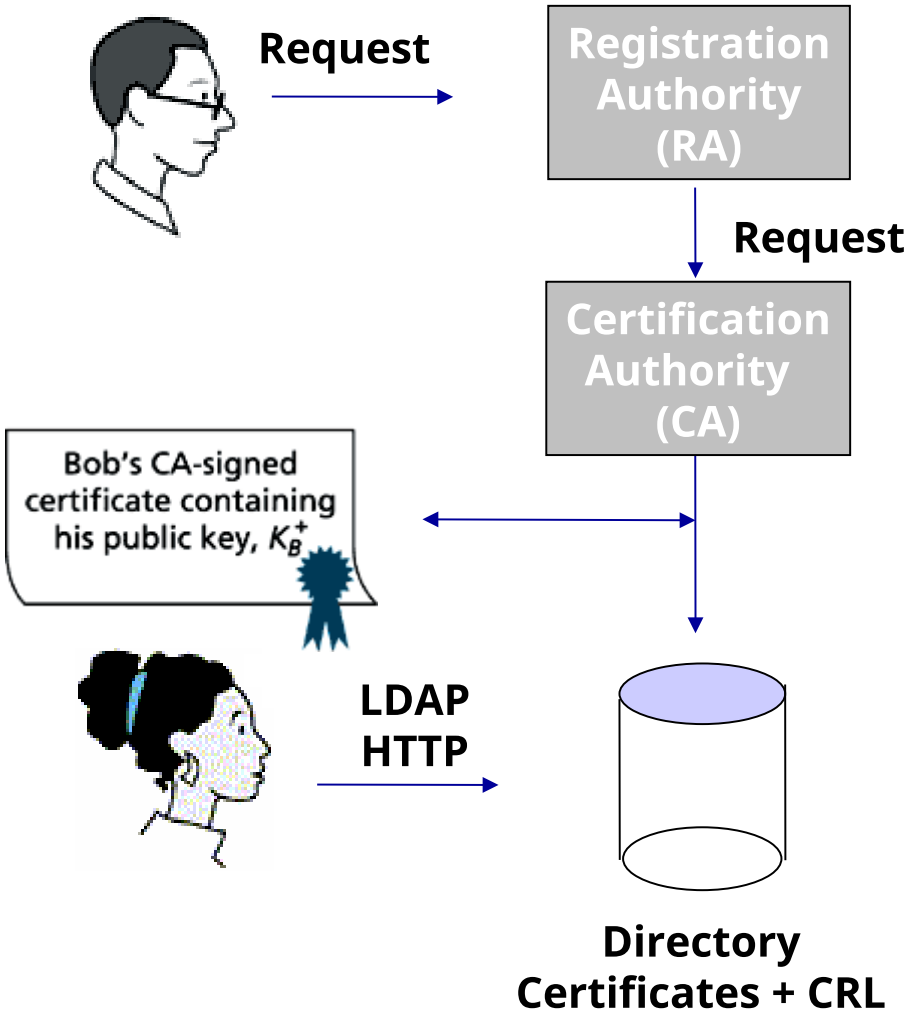
Key distribution and certification
Belgian eID Card

- Only for authentification and signature, not cyphering
- X.509 certificates stored on chip



	Private Key (Java Object)	Public Key (Java Object)	X.509 Certificates (Transparent file)
Basic	PrK#1		
Authentication	PrK#2	In Cert#2	Cert#2
Non-repudiation	PrK#3	In Cert#3	Cert#3
Certification Authority (CA)		In Cert#4	Cert#4
Commune		PuK#5	
Root			Cert#6
CA Role		PuK#7	
RN			Cert#8

Key distribution and certification
Public Key Infrastructure (PKI)



- PKI = authorities + directory + procedures (creation, update, revocation)
- Belgian eID
  - Publication of Certificate Revocation Lists (CRL)
  - Online Certificate Status Protocol (OCSP) service

Summary

- Computationally secure
- Cryptography
  - Symmetric key encryption: DES, AES
  - Public key encryption: RSA, ECC
  - Hash functions
- Key distribution and certification